

1. What is the Numeric Data Type in Python?

The Numeric data type is used to store numerical values. Python provides three numeric data types:

- **int** – Stores whole numbers.
- **float** – Stores decimal numbers.
- **complex** – Stores complex numbers (real + imaginary).

Example

```
a = 10    # int
```

```
b = 10.5  # float
```

```
c = 3 + 4j # complex
```

```
print(type(a))
```

```
print(type(b))
```

```
print(type(c))
```

2. What are Sequence Data Types in Python?

Sequence data types store multiple values in a specific order. Python provides three sequence data types:

- **str (String)** – Stores text.
- **list** – Stores multiple values and can be modified.
- **tuple** – Stores multiple values but cannot be modified.

Example

```
name = "Python"    # String
```

```
numbers = [10, 20, 30] # List
```

```
marks = (90, 95, 98)  # Tuple
```

3. What is the Set Data Type in Python?

A Set is an unordered collection of unique elements. It does not allow duplicate values. Python provides two set types:

- **set – Mutable.**
- **frozenset – Immutable.**

Example

```
numbers = {10, 20, 30, 20}
```

```
print(numbers)
```

Output

```
{10, 20, 30}
```

Duplicate values are automatically removed.

4. What is the Mapping Data Type in Python?

A Dictionary (dict) is a mapping data type that stores data as key-value pairs. Each key must be unique, and each key is associated with a value.

Example

```
student = {
```

```
    "name": "Thanuja",
```

```
    "age": 21,
```

```
    "branch": "AI & DS"
```

```
}
```

```
print(student["name"])
```

5. What is the Boolean Data Type in Python?

The **Boolean** data type represents only two values:

- **True**
- **False**

It is commonly used in conditions, comparisons, and decision-making statements.

Example

```
a = 10
```

```
b = 20
```

```
print(a < b)
```

6. What is the None Type in Python?

NoneType represents the absence of a value or a null value. It has only one value, **None**.

Example

```
x = None
```

```
print(type(x))
```